Supplier Reliability & Cost Analysis

Power BI Portfolio Project

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Tools: Power BI • DAX • Excel • Python

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This report reveals how supplier delivery performance impacts operational efficiency and cost.

It provides a clear decision-making framework to:

• Identify underperforming suppliers

• Quantify the financial impact of delays

• Understand root causes behind performance issues

• Prioritize improvement actions for maximum value

By combining operational and financial analytics,
this solution helps procurement and logistics teams make **smart, data-driven improvements**.

Business Question
Which suppliers deliver reliably and how fast?

Key Insight
We identify top and bottom performers based on OTD% and Lead Time.

Problem Frequency (Delay %)

0,26

Speed of Delivery - Avg Lead Time (days)

12,82

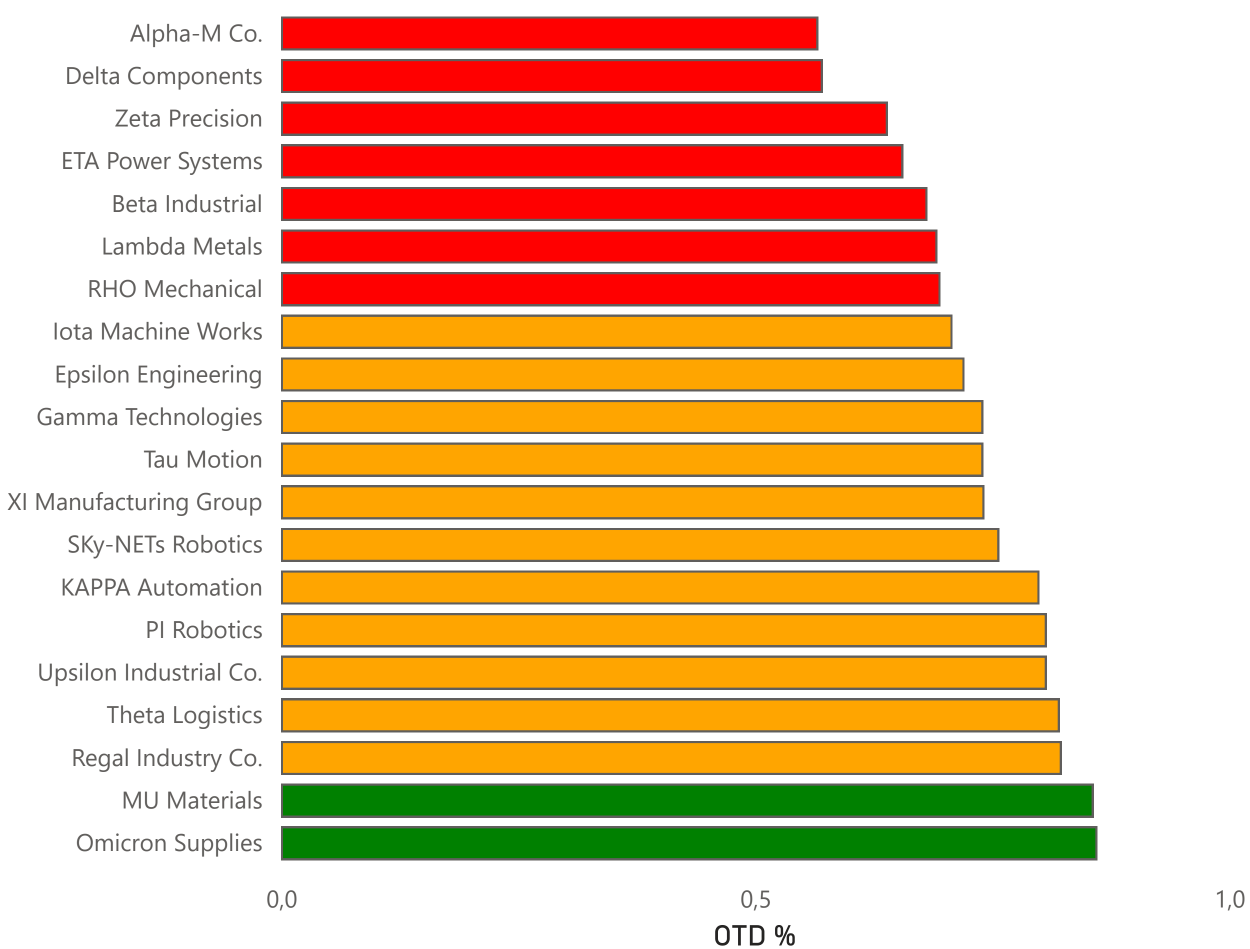
Supplier Reliability (OTD %)

0,74

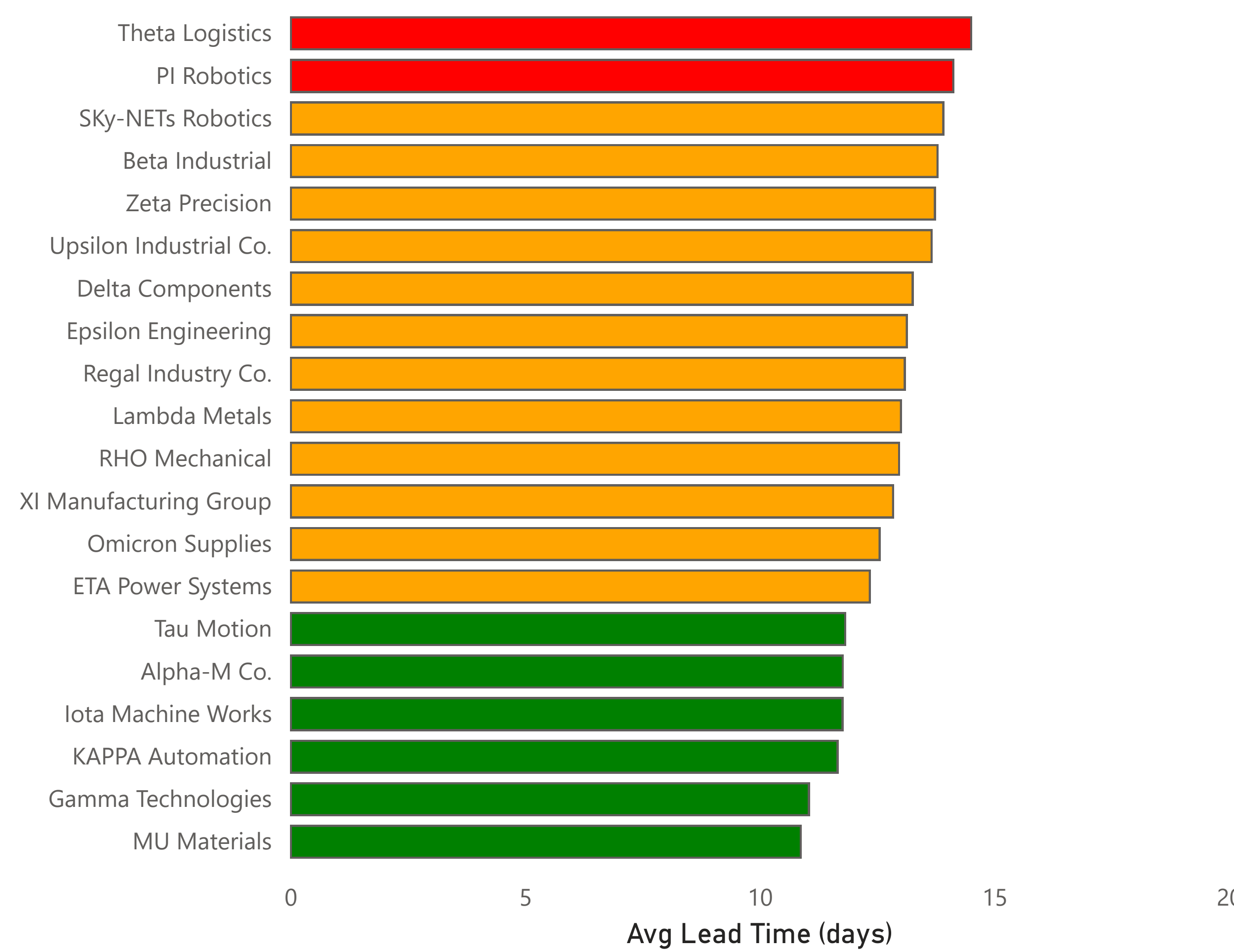
COUNTRY

All

Who is failing on-time delivery?



Who delivers too slowly?



Total Delay Cost (€)

82,7K

Avg Cost per Delayed Order (€)

537,01

Worst Supplier

Alpha-M Co.

Business Question

Which delays cost the most money?

Key Insight

Few suppliers create the majority of total delay cost.

Financial Impact Analysis – Page 4

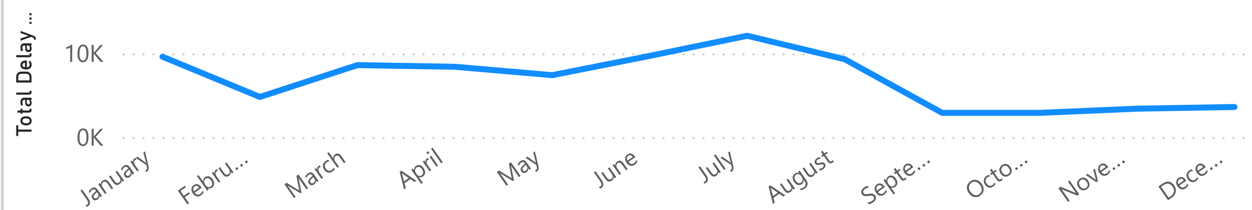
Country

All

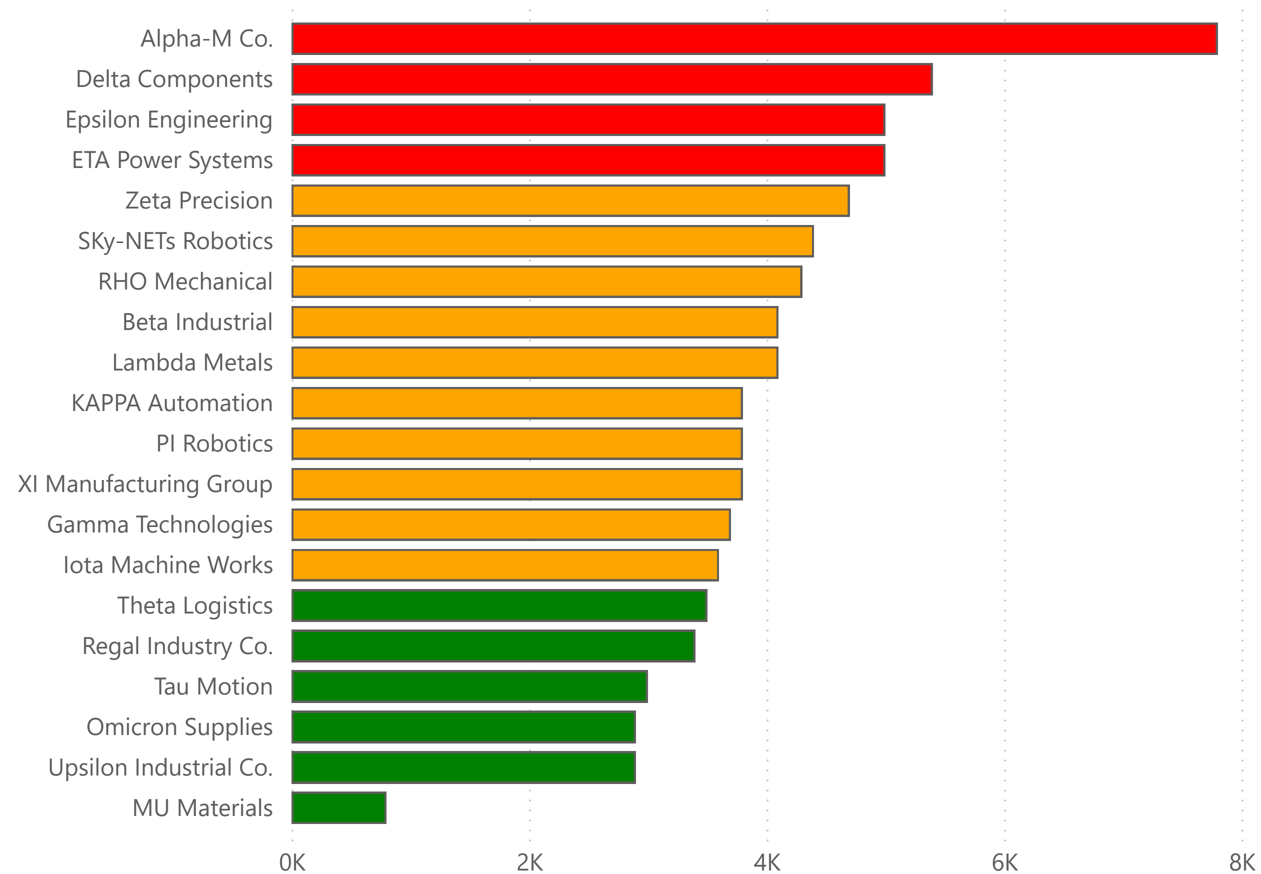
Key Insights

- Only 4 suppliers generate 30% of total delay costs — key priority.
- Documentation Error is the most expensive root cause → automation potential.
- Cost increased during May–Jul → supply coordination issues?
- Improving supplier performance could reduce losses by €20K–€30K per month.

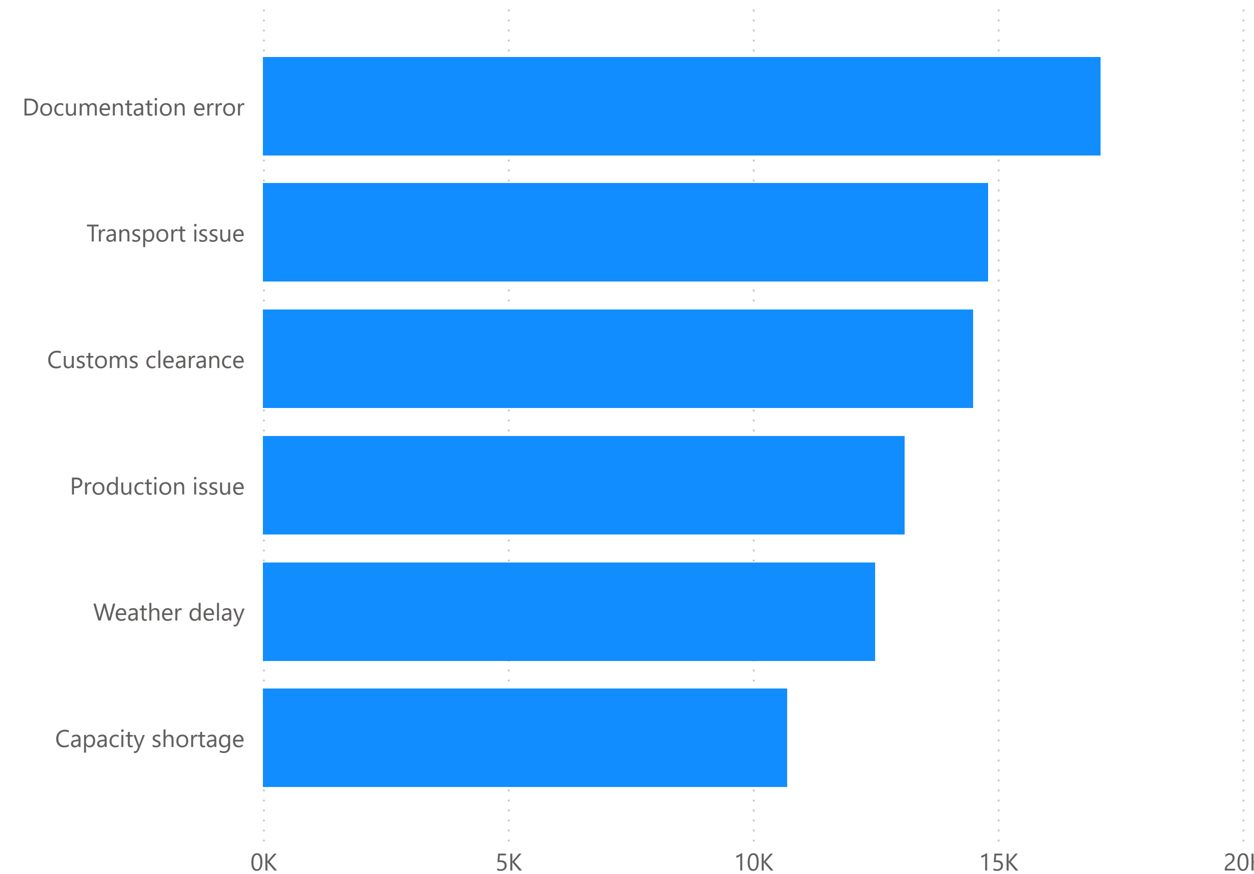
Are financial losses increasing over time?



Which suppliers create the biggest financial losses?



Which delay causes cost us the most?



Top Cost Cause (Reason)

Documentation error

Top Frequency Cause

Customs clearance

Business Question
Why do delays happen, and what should we fix first?

Key Insight
Documentation Errors are most expensive; Customs clearance are most frequent.

Recommended Actions:

• Only a few suppliers should be prioritized for performance improvement.
• The highest cost driver is Documentation Error → digitize/automate documents.
• High Frequency delays → focus on early warning signals for Customs clearance.
• Introduce supplier scorecards to reduce repeated issues.

Risk Summary

• Most suppliers are classified as Low Risk — overall stable delivery performance.
• Four suppliers have total delay cost over 5k → priority for corrective actions.
• Frequent delays from Customs clearance indicate need for monitoring and process improvements.
• Documentation Errors drive the highest cost → automation and standardization recommended.

Where do suppliers fail and why?

SupplierName	Capacity shortage	Customs clearance	Documentation error	Production issue	Transport issue	Weather dela
Alpha-M Co.	1200	1600	600	1300	2800	30
Delta Components	1100	300	900	900	500	170
Epsilon Engineering	300	500	2400	1200		60
ETA Power Systems	700	1800	1200	700	200	40
Zeta Precision	200	400	200	1000	2100	80
SKy-NETs Robotics	900	100	1500	600	800	50
RHO Mechanical	700	800		900	1200	70
Beta Industrial		1400		1500	200	100
Lambda Metals		1600	2100		200	20
KAPPA Automation	100	300	2500		300	60
PI Robotics		1300	500	1000	1000	
XI Manufacturing Group	1400	700		100	800	80
Gamma Technologies	700	100	800	200	600	130
Iota Machine Works		700	800		700	140
Theta Logistics	800	1100	1000		600	
Regal Industry Co.	1100		1100		1200	
Tau Motion	1400	200		300	1100	
Omicron Supplies			800	600	500	100
Upsilon Industrial Co.		700	600	1600		
MU Materials	100		100	600		

Which suppliers require action first based on delay cost & frequency?